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2027 CYCLE

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CUET UG

Applied Mathematics

Probability Distributions

Discrete & Continuous Variables, Expectation, Variance, Binomial, Poisson & Normal Models · Section B2 Unit V

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment & Exam Blueprint

In the NTA CUET (UG) Applied Mathematics examination, **Probability Distributions** (Unit V of Section B2) contributes **6 to 8 questions** (30 to 40 marks). The unit evaluates a student's proficiency in analyzing random processes: probability mass functions, mathematical expectations, and the three canonical distribution archetypes (Binomial, Poisson, and Normal).

Sub-Topic	Official NTA Syllabus Coverage	High-Yield CUET Focus
Random Variables & PMF	Discrete and continuous random variables; probability mass function (PMF); probability density function (PDF); mathematical expectation ($E(X)$) and variance ($\text{Var}(X)$).	Solving for normalization constant k from $\sum p_i = 1$; evaluating $E(X) = \sum x_i p_i$; variance identity $\text{Var}(X) = E(X^2) - [E(X)]^2$.
Linear Transformations	Expectation and variance of linear combinations; $E(aX+b) = aE(X) + b$; $\text{Var}(aX+b) = a^2\ \text{Var}(X)$.	Scaling and shift invariant properties; constant shift does NOT alter variance ($\text{Var}(X+c) = \text{Var}(X)$).
Binomial Distribution	Bernoulli trials; conditions for binomial model; PMF $P(X=r) = \binom{n}{r}p^r q^{n-r}$; mean np; variance npq; finding parameters from moments.	Mean strictly exceeds variance ($np > npq$); unimodal vs bimodal conditions $(n+1)p$; success/failure complement $p+q=1$.
Poisson Distribution	Poisson distribution as a limiting form of binomial distribution; rare events; PMF $P(X=r) = e^{-\lambda}\frac{\lambda^r}{r!}$; mean and variance equality.	Defining signature: Mean = Variance = λ; zero event probability $P(0) = e^{-\lambda}$; recurrence relation $P(r+1) = \frac{\lambda}{r+1}P(r)$.
Normal Distribution	Continuous Gaussian distribution; PDF; properties of normal curve; standard normal transformation $Z = \frac{X-\mu}{\sigma}$; empirical 68-95-99.7% rule.	Bell curve symmetry about μ; mean = median = mode; points of inflection $\mu \pm \sigma$; area calculations using standard normal Z-table.

CUET EXAM SCORING STRATEGY — PROBABILITY DISTRIBUTIONS

Distribution questions in CUET can be identified and solved rapidly by inspecting variance-to-mean ratios:

1. **Binomial:** Variance < Mean (since $npq < np$).
2. **Poisson:** Variance = Mean = λ.
3. **Normal:** Continuous distribution; exact point probability is ALWAYS zero ($P(X=c) = 0$).

Master these three signatures to immediately eliminate distractor options.

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1. Random Variables, Expectation & Variance

Discrete vs Continuous Random Variables

A **random variable (X)** is a real-valued mathematical function defined on the sample space S of a random experiment, assigning a numerical value to each experimental outcome:

1. **Discrete Random Variable:** Takes isolated, countable values (e.g., number of defective bulbs in a carton: $X \in \{0, 1, 2, ..., n\}$).
   - Governed by a **Probability Mass Function (PMF)**: $P(X = x_i) = p_i$.
   - **Axioms:** $p_i \geq 0$ for all i, and $\sum_i p_i = 1$.
2. **Continuous Random Variable:** Takes any real value in an unbroken numerical interval $[a, b] \subseteq \mathbb{R}$ (e.g., battery lifespan, human height, reaction time).
   - Governed by a **Probability Density Function (PDF)**: $f(x)$.
   - **Axioms:** $f(x) \geq 0$ for all x, and $\int_{-\infty}^{\infty} f(x)dx = 1$.
   - Probability of an interval: $P(a \leq X \leq b) = \int_a^b f(x)dx$.
   - For any single isolated point: $P(X = c) = \int_c^c f(x)dx = 0$.

Mathematical Expectation (Mean) and Variance

EXPECTATION & VARIANCE FORMULAS (DISCRETE CASE)

1. **Mean / Expected Value (μ or $E(X)$):**
$$E(X) = \mu = \sum_i x_i p_i$$
2. **Expectation of a Function $g(X)$:**
$$E[g(X)] = \sum_i g(x_i) p_i$$
3. **Second Moment about Origin:**
$$E(X^2) = \sum_i x_i^2 p_i$$
4. **Variance (σ^2 or $\text{Var}(X)$):**
$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2] = E(X^2) - [E(X)]^2 = \sum_i x_i^2 p_i - \mu^2$$
5. **Standard Deviation (σ):**
$$\sigma = \sqrt{\text{Var}(X)}$$

Linear Transformation Theorems

For any constants a and b:

1. $E(aX + b) = aE(X) + b$
2. $E(aX) = aE(X)$ and $E(b) = b$
3. $\text{Var}(aX + b) = a^2\ \text{Var}(X)$
4. $\text{Var}(aX) = a^2\ \text{Var}(X)$
5. $\text{Var}(X + b) = \text{Var}(X)$ (Shifting data by constant b does NOT alter dispersion)
6. $\text{Var}(b) = 0$ (The variance of a constant is strictly zero)

WORKED EXAMPLE 1 — NORMALIZATION CONSTANT AND EXPECTATION

A discrete random variable X has the following probability distribution:

x_i	0	1	2	3
$P(X=x_i)$	k	$3k$	$5k$	$7k$

Find (1) the value of k, (2) $E(X)$, and (3) $\mathrm{Var}(X)$.

Solution:

1. Normalization axiom: $\sum p_i = 1$.

$$k + 3k + 5k + 7k = 16k = 1 \Longrightarrow k = \frac{1}{16}.$$

2. Mathematical Expectation $E(X)$:

$$E(X) = \sum x_i p_i = (0)(k) + (1)(3k) + (2)(5k) + (3)(7k) = (3 + 10 + 21)k = 34k = \frac{34}{16} = \frac{17}{8} = 2.125.$$

3. Second Moment $E(X^2)$:

$$E(X^2) = \sum x_i^2 p_i = (0^2)(k) + (1^2)(3k) + (2^2)(5k) + (3^2)(7k) = (3 + 20 + 63)k = 86k = \frac{86}{16} = \frac{43}{8} = 5.375.$$

4. Variance:

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{43}{8} - \left(\frac{17}{8}\right)^2 = \frac{43}{8} - \frac{289}{64} = \frac{344 - 289}{64} = \frac{55}{64} \approx 0.8594.$$

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Cumulative Distribution Function (CDF)

For any random variable X (discrete or continuous), the **Cumulative Distribution Function (CDF)**, denoted $F(x)$, expresses the probability that X takes a value less than or equal to a specified threshold x:

$$F(x) = P(X \le x) = \begin{cases} \sum_{x_i \le x} p_i & \text{for discrete random variable} \\ \int_{-\infty}^{x} f(t)dt & \text{for continuous random variable} \end{cases}$$

Fundamental Properties of the CDF:

1. $0 \le F(x) \le 1$ for all real x.
2. Monotonically Non-Decreasing: If $x_1 < x_2$, then $F(x_1) \le F(x_2)$.
3. Boundary Limits: $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to +\infty} F(x) = 1$.
4. Interval Probability: $P(a < X \le b) = F(b) - F(a)$.
5. Relationship to PDF: For a continuous random variable, $f(x) = \frac{d}{dx}[F(x)]$.

WORKED EXAMPLE — CONTINUOUS DENSITY & EXPECTATION

A continuous random variable X has the probability density function:

$$f(x) = \begin{cases} kx(2-x) & 0 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

Determine: (1) the constant k, (2) the CDF $F(x)$ on $[0, 2]$, (3) $E(X)$, and (4) $P(X > 1)$.

Solution:

1. Total area normalization: $\int_0^2 f(x)dx = 1$.

$$k\int_0^2 (2x-x^2)dx = k\left[x^2 - \frac{x^3}{3}\right]_0^2 = k\left[4-\frac{8}{3}\right] = k\left(\frac{4}{3}\right) = 1 \Longrightarrow k = \frac{3}{4}.$$

2. Cumulative Distribution Function for $0 \le x \le 2$:

$$F(x) = \int_0^x \frac{3}{4}(2t-t^2)dt = \frac{3}{4}\left[t^2 - \frac{t^3}{3}\right]_0^x = \frac{3}{4}\left(x^2 - \frac{x^3}{3}\right) = \frac{3x^2-x^3}{4}.$$

3. Mathematical Expectation:

$$E(X) = \int_0^2 xf(x)dx = \frac{3}{4}\int_0^2 (2x^2-x^3)dx = \frac{3}{4}\left[\frac{2}{3}x^3 - \frac{x^4}{4}\right]_0^2 = \frac{3}{4}\left[\frac{16}{3}-\frac{16}{4}\right] = \frac{3}{4}\left[\frac{16}{3}-4\right] = \frac{3}{4}\left(\frac{4}{3}\right) = 1.0.$$

(Notice that $E(X) = 1.0$ matches the center of symmetry of the parabolic density).

4. $P(X > 1) = 1 - F(1) = 1 - \frac{3(1)^2-1^3}{4} = 1 - \frac{2}{4} = 1 - 0.5 = 0.5$.

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2. The Binomial Distribution

Bernoulli Trials and Derivation

A sequence of trials is termed **Bernoulli trials** if:

1. The number of trials n is finite and fixed in advance.
2. Each trial has exactly two mutually exclusive outcomes: **Success (S)** or **Failure (F)**.
3. The probability of success $P(S) = p$ remains constant from trial to trial ($0 < p < 1$), and $P(F) = q = 1 - p$.
4. Successive trials are statistically independent.

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BINOMIAL DISTRIBUTION PMF & PARAMETERS

Let X denote the number of successes in n independent Bernoulli trials. Then X follows a Binomial distribution, denoted $X \sim B(n,p)$:

$$P(X=r) = \binom{n}{r} p^r q^{n-r} \quad (r = 0,1,2,...,n)$$

- **Mean:** $E(X) = np$
- **Variance:** $\mathrm{Var}(X) = npq$
- **Standard Deviation:** $\sigma = \sqrt{npq}$
- **Mean vs Variance Relationship:**

$$\mathrm{Var}(X) = (np)\cdot q < np = E(X) \quad (\text{since } \ q = 1-p < 1)$$

In any binomial distribution, the mean is strictly greater than the variance!

Mode of the Binomial Distribution

The mode is the value of r that maximizes $P(X = r)$. It depends on the value of $(n+1)p$:

1. If $(n+1)p$ is an integer m: The distribution is **bimodal** with modes at $r = m-1$ and $r = m$.
2. If $(n+1)p$ is NOT an integer: The distribution is **unimodal** with unique mode at $r = \lfloor (n+1)p \rfloor$ (the greatest integer less than or equal to $(n+1)p$).

WORKED EXAMPLE 2 — PARAMETER RECOVERY FROM MOMENTS

For a binomial distribution, the mean is 4 and the variance is 3. Determine: (1) the parameters n and p, (2) $P(X = 0)$, and (3) $P(X \geq 1)$.

Solution: Given $E(X) = np = 4$ and $\text{Var}(X) = npq = 3$. Divide variance by mean:

$$q = \frac{npq}{np} = \frac{3}{4} = 0.75.$$

$$p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4} = 0.25.$$

Find n:

$$np = 4 \Longrightarrow n\left(\frac{1}{4}\right) = 4 \Longrightarrow n = 16.$$

1. Parameters are $n = 16$ and $p = \frac{1}{4}$.
2. $P(X = 0) = \binom{16}{0}\left(\frac{1}{4}\right)^0\left(\frac{3}{4}\right)^{16} = \left(\frac{3}{4}\right)^{16}$.
3. $P(X \geq 1) = 1 - P(X = 0) = 1 - \left(\frac{3}{4}\right)^{16}$.

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WORKED EXAMPLE — DETERMINING THE MODE OF A BINOMIAL MODEL

A student takes a multiple-choice quiz of $n = 15$ questions, guessing randomly among 4 options per question ($p = \frac{1}{4}$). Find the most probable number of correct answers (the mode).

Solution: Evaluate $(n + 1)p$:

$$(n + 1)p = (15 + 1)\left(\frac{1}{4}\right) = 16 \times \frac{1}{4} = 4.$$

Since $(n + 1)p = 4$ is an integer m, the distribution is **bimodal** with modes occurring at:

$$r = m - 1 = 3 \quad \text{and} \quad r = m = 4.$$

The student is equally most likely to guess exactly 3 or 4 answers correctly:

$$P(X = 3) = \binom{15}{3}\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^{12} = 455 \cdot \left(\frac{3^{12}}{4^{15}}\right) \approx 0.2252$$

$$P(X = 4) = \binom{15}{4}\left(\frac{1}{4}\right)^4\left(\frac{3}{4}\right)^{11} = 1365 \cdot \left(\frac{3^{11}}{4^{15}}\right) = 455 \cdot \left(\frac{3^{12}}{4^{15}}\right) \approx 0.2252.$$

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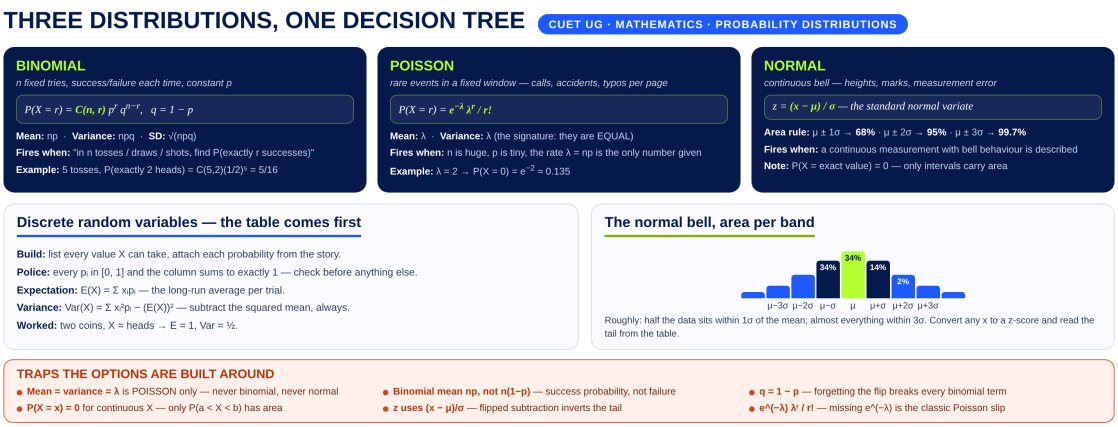


Figure 1: Figure 1: The Three Distribution Families Compared — Structural Parameters, Moments & Decision Trees.

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3. The Poisson Distribution

The Poisson Limit of the Binomial Distribution

The **Poisson distribution** was introduced by Siméon Denis Poisson (1837) as a limiting case of the Binomial distribution under the following conditions:

1. The number of trials n is indefinitely large ($n \to \infty$).
2. The probability of success p in each individual trial is extremely small ($p \to 0$, rare events).
3. The average number of successes remains a constant finite quantity:

$$\lambda = np > 0$$

POISSON DISTRIBUTION PMF & MOMENTS

If X represents the count of occurrences of a rare event, $X \sim \text{Poisson}(\lambda)$:

$$P(X = r) = \frac{e^{-\lambda} \cdot \lambda^r}{r!} \quad (r = 0, 1, 2, 3, \ldots)$$

where λ is the mean arrival/occurrence rate and $e \approx 2.71828$.

- **Mean:** $E(X) = \lambda$
- **Variance:** $\text{Var}(X) = \lambda$
- **Crucial Defining Property:**

$$\text{Mean} = \text{Variance} = \lambda$$

Whenever an exam problem notes that variance equals mean, the variable is Poisson!

Key Mathematical Properties of Poisson Distributions

1. **Recurrence Relation for Fast Computation:**

$$P(r+1) = \frac{\lambda}{r+1} \cdot P(r)$$

2. **Zero Probability Signature:**

$$P(X=0) = \frac{e^{-\lambda} \cdot \lambda^0}{0!} = e^{-\lambda}$$

$$P(X \geq 1) = 1 - P(X=0) = 1 - e^{-\lambda}$$

3. **Additive Property:** If $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$ are independent, then:

$$X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

WORKED EXAMPLE 3 — CUSTOMER SERVICE DESK MODELING

A call center receives incoming assistance calls at an average rate of $\lambda = 3$ calls per minute according to a Poisson process. Calculate: (1) the probability of receiving no calls in a minute, (2) the probability of receiving exactly 2 calls, and (3) the probability of at least one call. (Given: $e^{-3} \approx 0.0498$).

Solution: $X \sim \text{Poisson}(3)$ with $\lambda = 3$.

1. $P(X=0) = e^{-3} \approx 0.0498$ (or 4.98%).
2. $P(X=2) = \frac{e^{-3} \cdot 3^2}{2!} = \frac{0.0498 \times 9}{2} = \frac{0.4482}{2} = 0.2241$ (or 22.41%).
3. $P(X \geq 1) = 1 - P(X=0) = 1 - 0.0498 = 0.9502$ (or 95.02%).

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4. The Gaussian Normal Distribution

The Continuous Bell Curve Architecture

The **Normal (or Gaussian) Distribution** is the most important continuous distribution in applied mathematics. A continuous random variable X defined on $(-\infty, \infty)$ follows a Normal distribution with parameters μ (mean) and σ^2 (variance), denoted $X \sim N(\mu, \sigma^2)$, if its PDF is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (-\infty < x < \infty)$$

Geometric Property	Mathematical Statement & Examination Significance
Symmetry & Center	The curve is completely symmetrical and bell-shaped about the vertical ordinate $x = \mu$. Consequently: $$\text{Mean} = \text{Median} = \text{Mode} = \mu$$
Total Area & Half-Planes	Total area under the PDF curve is exactly 1.0: $$\int_{-\infty}^{\infty} f(x)dx = 1.0, \quad P(X \leq \mu) = P(X \geq \mu) = 0.50$$
Points of Inflection	The points where curvature changes from concave downward to concave upward occur at: $$x = \mu - \sigma \quad \text{and} \quad x = \mu + \sigma$$
Maximum Ordinate	The maximum peak of $f(x)$ occurs at $x = \mu$, where height is $y_{\max} = \frac{1}{\sigma\sqrt{2\pi}}$.
Asymptotic Tails	The tails extend indefinitely to $-\infty$ and $+\infty$, asymptotically approaching the horizontal axis without ever touching it.

Standard Normal Distribution (Z)

Any general normal variable $X \sim N(\mu, \sigma^2)$ can be transformed to the **Standard Normal Variable (Z)** via the standard score transformation:

$$Z = \frac{X - \mu}{\sigma}$$

The variable Z has mean $E(Z) = 0$ and standard deviation $\sigma_Z = 1$, denoted $Z \sim N(0, 1)$. Its PDF is:

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{z^2}{2}}$$

The Empirical Rule (68-95-99.7% Rule)

For any normal distribution, the probability of an observation falling within specified standard deviations of the mean is:

1. $P(\mu - \sigma \leq X \leq \mu + \sigma) = P(-1 \leq Z \leq 1) \approx 0.6827$ (68.27% of all data)
2. $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = P(-2 \leq Z \leq 2) \approx 0.9545$ (95.45% of all data)
3. $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) = P(-3 \leq Z \leq 3) \approx 0.9973$ (99.73% of all data)

QUARTILES & DISPERSION METRICS OF NORMAL DISTRIBUTION

- **First Quartile (Q_1):** $Q_1 = \mu - 0.6745\sigma$
- **Third Quartile (Q_3):** $Q_3 = \mu + 0.6745\sigma$
- **Quartile Deviation (QD):** $\text{QD} = \frac{Q_3 - Q_1}{2} = 0.6745\sigma \approx \frac{2}{3}\sigma$
- **Mean Deviation (MD):** $\text{MD} = \sqrt{\frac{2}{\pi}}\sigma \approx 0.7979\sigma \approx \frac{4}{5}\sigma$
- **Ratio Rule:** $\text{QD} : \text{MD} : \sigma = 10 : 12 : 15$ (since $\frac{2}{3} : \frac{4}{5} : 1 = 10 : 12 : 15$).

WORKED EXAMPLE 4 — STANDARD NORMAL PROBABILITY

The test scores in an entrance examination are normally distributed with mean $\mu = 500$ and standard deviation $\sigma = 100$. Find the percentage of candidates scoring above 650. (Given: For $Z = 1.5$, area from 0 to 1.5 is 0.4332).

Solution: Convert raw score $X = 650$ to standard normal Z:

$$Z = \frac{X - \mu}{\sigma} = \frac{650 - 500}{100} = \frac{150}{100} = 1.5.$$

We need $P(X > 650) = P(Z > 1.5)$. Using symmetry:

$$P(Z > 1.5) = P(Z > 0) - P(0 < Z < 1.5) = 0.5000 - 0.4332 = 0.0668 \quad \text{(or 6.68\%)}.$$

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Parameter Recovery from Normal Percentiles (Inverse Normal Problem)

In competitive CUET examinations, candidates are frequently given observed percentage thresholds and required to determine the underlying distribution parameters μ and σ.

WORKED EXAMPLE — SIMULTANEOUS PARAMETER RECOVERY

In an aptitude examination, 10% of candidates scored above 75 marks, while 20% scored below 40 marks. Assuming scores are normally distributed, determine the mean μ and standard deviation σ. (Given: For upper tail area 0.10, $Z = 1.28$; for lower tail area 0.20, $Z = -0.84$).

Solution:

1. Upper 10% threshold ($X_1 = 75$):

$$P(X > 75) = 0.10 \Longrightarrow Z_1 = \frac{75 - \mu}{\sigma} = +1.28 \Longrightarrow 75 - \mu = 1.28\sigma \quad ---(1)$$

2. Lower 20% threshold ($X_2 = 40$):

$$P(X < 40) = 0.20 \Longrightarrow Z_2 = \frac{40 - \mu}{\sigma} = -0.84 \Longrightarrow 40 - \mu = -0.84\sigma \quad ---(2)$$

3. Subtract equation (2) from equation (1):

$$(75 - \mu) - (40 - \mu) = 1.28\sigma - (-0.84\sigma)$$

$$35 = 2.12\sigma \Longrightarrow \sigma = \frac{35}{2.12} \approx 16.51 \text{ marks.}$$

4. Substitute $\sigma = 16.51$ back into equation (1):

$$\mu = 75 - 1.28(16.51) = 75 - 21.13 \approx 53.87 \text{ marks.}$$

The Normal Approximation to the Binomial Distribution

When the number of Bernoulli trials n is large and the probability of success p is moderately balanced (neither p nor q close to zero):

$$np \geq 5 \quad \text{and} \quad nq \geq 5$$

the discrete binomial distribution $B(n, p)$ can be closely approximated by a continuous normal distribution:

$$X_{\text{approx}} s\,\text{im}\, N(\mu = np, \sigma^2 = npq)$$

Continuity Correction: Because a continuous distribution replaces a discrete variable, an integer count $X = r$ spans the continuous interval $[r - 0.5, r + 0.5]$:

- $P(X = r) \approx P(r - 0.5 \leq X_{\text{norm}} \leq r + 0.5)$
- $P(X \geq r) \approx P(X_{\text{norm}} \geq r - 0.5)$
- $P(X \leq r) \approx P(X_{\text{norm}} \leq r + 0.5)$

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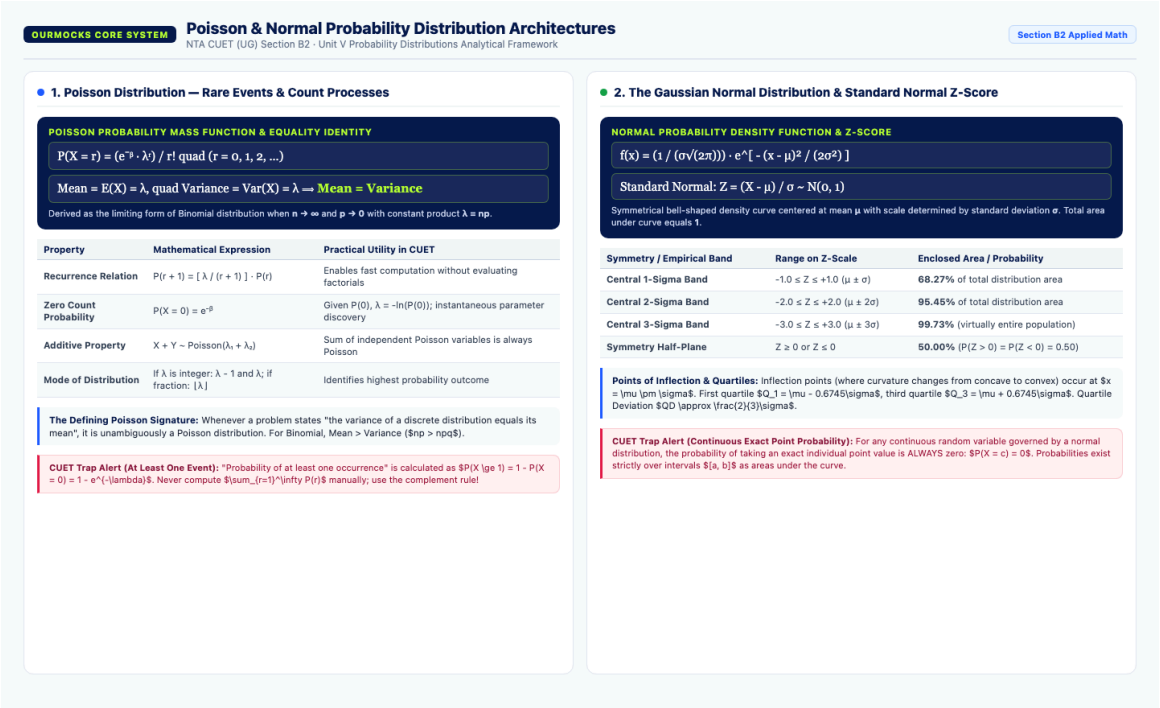


Figure 2: Figure 2: Poisson Process Modeling, Gaussian Bell Curve Geometry & The Standard Normal Z-Score Transformation.

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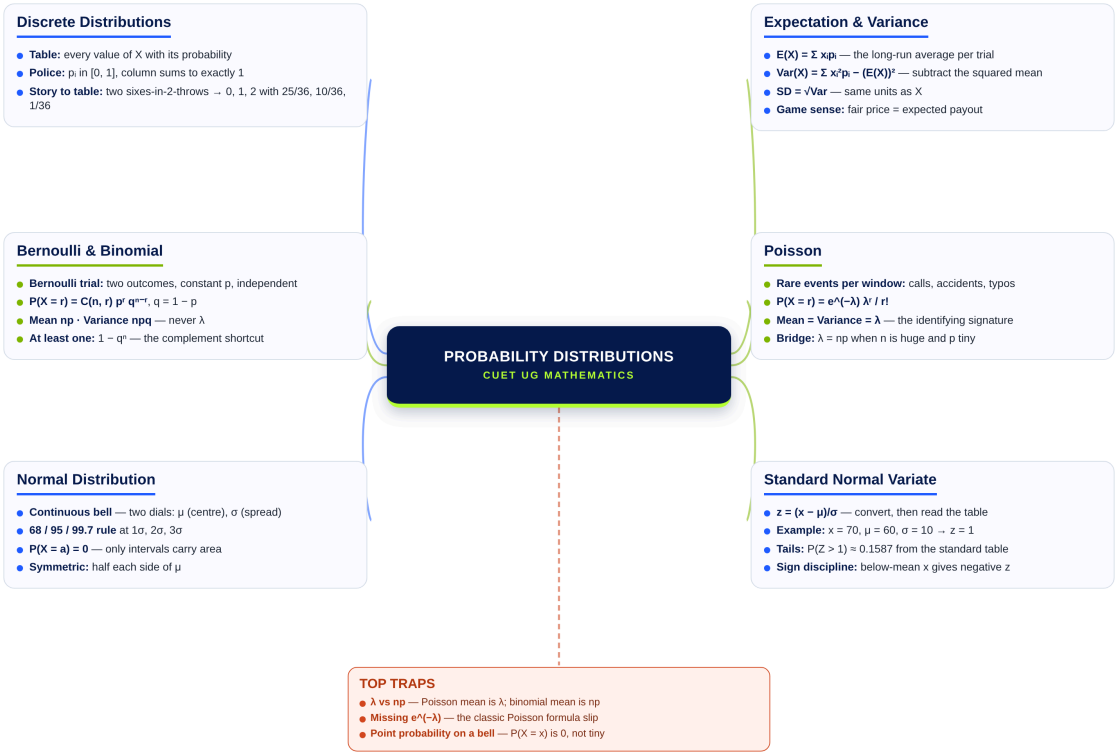


Figure 3: Figure 3: Comprehensive Probability Distributions Mindmap — Interconnecting Discrete PMF & Continuous PDF Architectures.

HIGH-YIELD MEMORY HOOKS FOR CUET 2027

Keep these fundamental distribution signatures primed:

- **Discrete Normalization:** $\sum p_i = 1$; Expectation $E(X) = \sum x_i p_i$.
- **Variance Identity:** $\text{Var}(X) = E(X^2) - [E(X)]^2$.
- **Scaling:** $\text{Var}(aX + b) = a^2\ \text{Var}(X)$. Constant shifts never change variance!
- **Binomial Signature:** Mean $= np >$ Variance $= npq$.
- **Poisson Signature:** Mean = Variance $= \lambda$.
- **Zero Poisson:** $P(X = 0) = e^{-\lambda} \implies P(X \geq 1) = 1 - e^{-\lambda}$.
- **Normal Geometry:** Bell curve is symmetric about μ; $P(X = c) = 0$ for single points.
- **Empirical Bands:** 68.27% in $\pm 1\sigma$, 95.45% in $\pm 2\sigma$, 99.73% in $\pm 3\sigma$.

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MASTER FORMULA & DECISION CHEATSHEET — PROBABILITY DISTRIBUTIONS

1. Moments & Linear Transformations:

$$E(X) = \sum x_i p_i, \qquad \text{Var}(X) = \sum x_i^2 p_i - \mu^2$$

$$E(aX + b) = aE(X) + b, \qquad \text{Var}(aX + b) = a^2 \text{ Var}(X)$$

2. Binomial Model $B(n, p)$:

$$P(X = r) = \binom{n}{r} p^r q^{n-r}, \qquad \mu = np, \qquad \sigma^2 = npq < np$$

3. Poisson Model Poisson(λ):

$$P(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}, \qquad \mu = \lambda, \qquad \sigma^2 = \lambda = \mu$$

$$P(r + 1) = \frac{\lambda}{r + 1} P(r), \qquad P(X \geq 1) = 1 - e^{-\lambda}$$

4. Gaussian Normal Model $N(\mu, \sigma^2)$:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1), \qquad P(\mu - \sigma \leq X \leq \mu + \sigma) = 68.27\%$$

$$\text{QD} \approx \frac{2}{3}\sigma, \qquad \text{MD} \approx \frac{4}{5}\sigma, \qquad \text{QD} : \text{MD} : \sigma = 10 : 12 : 15$$

THE SEVEN FATAL CUET TRAPS IN PROBABILITY DISTRIBUTIONS

1. **The Variance Constant Shift Blindspot:** Calculating $\text{Var}(3X + 5) = 9 \text{ Var}(X) + 5$ or $9 \text{ Var}(X) + 25$. Remember: shifting data by $+5$ leaves spread unchanged! $\text{Var}(aX + b) = a^2 \text{ Var}(X)$.
2. **Negative Variance Disaster:** Forgetting that $\text{Var}(X) = E(X^2) - [E(X)]^2$ must strictly be non-negative (≥ 0). If your result is negative, you subtracted in the wrong order!
3. **Binomial Variance Exceeding Mean Fallacy:** Choosing a binomial model when variance exceeds mean ($\sigma^2 > \mu$). That is physically impossible because $q = 1 - p < 1$.
4. **Poisson Mean-Variance Inconsistency:** Searching for two separate parameters for a Poisson distribution. The single parameter λ serves simultaneously as **both** mean and variance!
5. **Normal Exact Point Probability Fallacy:** Calculating $P(X = 50)$ for a continuous normal variable. In any continuous distribution, the probability at any isolated single point is strictly zero ($P(X = c) = 0$).
6. **Empirical Rule Half-Plane Confusion:** When asked for $P(X > \mu + \sigma)$, candidates often write 68.27%. The correct answer is $\frac{100\% - 68.27\%}{2} = 15.865\%$!
7. **Poisson "At Least One" Exponent Slip:** Calculating $P(X \geq 1)$ as $e^{-\lambda}$. Remember: $e^{-\lambda} = P(X = 0)$; therefore $P(X \geq 1) = 1 - e^{-\lambda}$!

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Practice Questions — Part 1 (Q1 to Q5)

Q1. A discrete random variable X has the probability distribution: $P(X=-1)=0.2$, $P(X=0)=0.5$, $P(X=1)=0.3$. The mathematical expectation $E(X)$ is:

(A) 0.1 (B) 0.0
(C) 0.5 (D) 0.3

Sol. Q1. The expectation of a discrete random variable is given by:

$$E(X)=\sum x_i p_i$$

Substitute the given values:

$$E(X)=(-1)(0.2)+(0)(0.5)+(1)(0.3)=-0.2+0+0.3=+0.1.$$

Matches Option A. **Answer: (A)**

Q2. If X is a random variable with mean $E(X)=5$ and variance $\mathrm{Var}(X)=4$, then $E(2X+3)$ and $\mathrm{Var}(2X+3)$ are respectively:

(A) 13 and 11 (B) 13 and 16
(C) 10 and 16 (D) 13 and 19

Sol. Q2. By the linear transformation theorems of expectation and variance:

1. $E(2X+3)=2E(X)+3=2(5)+3=10+3=13$.
2. $\mathrm{Var}(2X+3)=2^2\cdot\mathrm{Var}(X)=4\cdot 4=16$.

(Notice that adding the constant 3 does not alter variance: $\mathrm{Var}(aX+b)=a^2\,\mathrm{Var}(X)$). Matches Option B. **Answer: (B)**

Q3. For a certain binomial distribution, the mean is 6 and the variance is 2. The total number of trials n is:

(A) 9 (B) 12
(C) 8 (D) 18

Sol. Q3. For a Binomial distribution $B(n,p)$:

- Mean $\mu=np=6$.
- Variance $\sigma^2=npq=2$.

Divide variance by mean:

$$q=\frac{npq}{np}=\frac{2}{6}=\frac{1}{3}.$$

Therefore, success probability $p=1-q=1-\frac{1}{3}=\frac{2}{3}$. Now substitute $p=\frac{2}{3}$ into the mean equation:

$$np=6\Longrightarrow n\left(\frac{2}{3}\right)=6\Longrightarrow n=6\times\frac{3}{2}=9.$$

Matches Option A. **Answer: (A)**

Q4. Which of the following conditions is an absolute requirement for a random variable X to follow a Poisson distribution?

(A) The variance must strictly exceed the mean (B) The variance must be strictly equal to the mean
(C) The number of trials must be small ($n<30$) (D) The probability of success must be greater than 0.5

Sol. Q4. The hallmark theoretical property of the Poisson distribution is that its variance is strictly identical to its mean: $\mathrm{Var}(X)=E(X)=\lambda$. In a Binomial distribution, variance is strictly less than mean ($npq<np$), while in a Negative Binomial distribution, variance exceeds mean. **Answer: (B)**

Q5. If X follows a Poisson distribution such that $P(X=1)=P(X=2)$, the parameter λ of the distribution is:

(A) 1 (B) 2
(C) 0.5 (D) 4

Sol. Q5. The Poisson PMF is $P(X=r)=\frac{e^{-\lambda}\lambda^r}{r!}$. Given $P(X=1)=P(X=2)$:

$$\frac{e^{-\lambda}\lambda^1}{1!}=\frac{e^{-\lambda}\lambda^2}{2!}$$

Divide both sides by $e^{-\lambda}\lambda$ (since $\lambda>0$):

$$\frac{1}{1}=\frac{\lambda}{2}\Longrightarrow\lambda=2.$$

Matches Option B. **Answer: (B)**

Practice Questions — Part 2 (Q6 to Q10)

Q6. For a continuous random variable X following a normal distribution $N(\mu, \sigma^2)$, the probability that X takes the exact single point value $X = \mu$ is:

(A) 0.5 (B) 1.0
(C) 0 (D) $\frac{1}{\sigma\sqrt{2\pi}}$

Sol. Q6. For any continuous random variable governed by a continuous probability density function $f(x)$, probability represents the geometric area under the curve. Because a single point has zero width ($dx = 0$), the integral evaluated at an isolated point is strictly zero: $P(X = \mu) = \int_{\mu}^{\mu} f(x)dx = 0$. (Option D is the height of the density ordinate, NOT probability). **Answer: (C)**

Q7. The heights of soldiers in a regiment are normally distributed with mean $\mu = 175$ cm and standard deviation $\sigma = 5$ cm. What percentage of soldiers have heights between 165 cm and 185 cm?

(A) 68.27% (B) 95.45%
(C) 99.73% (D) 50.00%

Sol. Q7. Transform the boundary values to standard normal Z-scores:

- For $X = 165$: $Z_1 = \frac{165-175}{5} = -\frac{10}{5} = -2.0$.
- For $X = 185$: $Z_2 = \frac{185-175}{5} = +\frac{10}{5} = +2.0$.

The interval $[165, 185]$ corresponds to $\mu \pm 2\sigma$ (from $Z = -2$ to $Z = +2$). By the Empirical Rule of the normal distribution, the area enclosed within $\pm 2\sigma$ is precisely 95.45%. **Answer: (B)**

Q8. If a random variable X has $E(X) = 3$ and $E(X^2) = 13$, the standard deviation of X is:

(A) 10 (B) 4
(C) 2 (D) $\sqrt{10}$

Sol. Q8. The variance is given by:

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 13 - (3)^2 = 13 - 9 = 4.$$

The standard deviation is the positive square root of variance:

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{4} = 2.$$

Matches Option C. **Answer: (C)**

Q9. In a family with 4 children, assuming boys and girls are equally likely ($p = 0.5$), the probability that the family has at least one boy is:

(A) $\frac{15}{16}$ (B) $\frac{1}{16}$
(C) $\frac{7}{8}$ (D) $\frac{3}{4}$

Sol. Q9. Here $n = 4$ independent trials with $p = P(\text{Boy}) = 0.5$ and $q = P(\text{Girl}) = 0.5$. $P(\text{At least one boy}) = 1 - P(\text{No boys}) = 1 - P(X = 0)$.

$$P(X = 0) = \binom{4}{0}\left(\frac{1}{2}\right)^0\left(\frac{1}{2}\right)^4 = \frac{1}{16}.$$

$$P(X \geq 1) = 1 - \frac{1}{16} = \frac{15}{16}.$$

Matches Option A. **Answer: (A)**

Q10. If X is a Poisson random variable with parameter $\lambda = 4$, the mode of the distribution is:

(A) 4 only (B) 3 only
(C) Both 3 and 4 (D) 2

Sol. Q10. For a Poisson distribution, if the parameter λ is a positive integer, the distribution is **bimodal** with modes occurring at two consecutive integers:

$$r = \lambda - 1 \quad \text{and} \quad r = \lambda.$$

Here $\lambda = 4$ is an integer. Thus, the modes are $4 - 1 = 3$ and 4. (Check: $P(3) = \frac{e^{-4}4^3}{6} = \left(\frac{64}{6}\right)e^{-4} = 10.67e^{-4}$; and $P(4) = \frac{e^{-4}4^4}{24} = \left(\frac{256}{24}\right)e^{-4} = 10.67e^{-4}$). **Answer: (C)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. For a normal distribution with mean $\mu = 50$ and variance $\sigma^2 = 25$, the two points of inflection of the curve occur at:

(A) 45 and 55 **(B)** 40 and 60
(C) 47.5 and 52.5 **(D)** 25 and 75

Sol. Q11. Given $\mu = 50$ and $\sigma^2 = 25 \Rightarrow \sigma = \sqrt{25} = 5$. In a normal distribution, the points of inflection (where the second derivative $f''(x) = 0$ and the curve transitions from concave downward to concave upward) occur at exactly one standard deviation from the mean:

$$x = \mu - \sigma = 50 - 5 = 45$$
$$x = \mu + \sigma = 50 + 5 = 55$$

The points of inflection are at 45 and 55. Matches Option A. **Answer: (A)**

Q12. A factory produces light bulbs with a defect rate of 1%. In a random sample of 200 bulbs, using the Poisson approximation to the Binomial distribution, the average number of defective bulbs λ is:

(A) 1 **(B)** 2
(C) 0.02 **(D)** 4

Sol. Q12. When sample size n is large and success probability p is small, the Poisson distribution approximates the Binomial distribution with parameter:

$$\lambda = np$$

Given $n = 200$ and $p = 1\% = 0.01$:

$$\lambda = 200 \times 0.01 = 2.$$

Matches Option B. **Answer: (B)**

Q13. If the Quartile Deviation of a normally distributed random variable is 10, the estimated standard deviation σ of the distribution is approximately:

(A) 15 **(B)** 20
(C) 6.67 **(D)** 12

Sol. Q13. For a normal distribution, the relationship between Quartile Deviation (QD) and standard deviation (σ) is:

$$\text{QD} \approx \frac{2}{3}\sigma$$

Given QD = 10:

$$10 = \frac{2}{3}\sigma \Rightarrow \sigma = 10 \times \frac{3}{2} = 15.$$

Matches Option A. **Answer: (A)**

Q14. Two independent random variables X and Y follow Poisson distributions with parameters $\lambda_1 = 2$ and $\lambda_2 = 3$ respectively. The distribution of the sum $W = X + Y$ is:

(A) Poisson(6) **(B)** Poisson(5)
(C) Binomial$(5, 0.5)$ **(D)** $N(5, 5)$

Sol. Q14. By the additive (reproductive) property of the Poisson distribution: If $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ are independent random variables, their sum $X + Y$ also follows a Poisson distribution whose parameter is the sum of the individual parameters:

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2) = \text{Poisson}(2 + 3) = \text{Poisson}(5).$$

Matches Option B. **Answer: (B)**

Q15. In an eight-trial binomial experiment $B(8, p)$, if the probability of 3 successes equals the probability of 5 successes ($P(X = 3) = P(X = 5)$), the probability of success p is:

(A) 0.5 **(B)** 0.375
(C) 0.625 **(D)** 0.25

Sol. Q15. The Binomial PMF is $P(X = r) = \binom{n}{r} p^r q^{n-r}$. Given $n = 8$ and $P(X = 3) = P(X = 5)$:

$$\binom{8}{3} p^3 q^{8-3} = \binom{8}{5} p^5 q^{8-5}$$

Since $\binom{8}{3} = \binom{8}{5} = 56$:

$$56p^3q^5 = 56p^5q^3$$

Divide both sides by $56p^3q^3$ (assuming $0 < p, q < 1$):

$$q^2 = p^2 \Rightarrow q = p$$

Since $p + q = 1$, we have $p + p = 1 \Rightarrow 2p = 1 \Rightarrow p = 0.5$. Matches Option A. **Answer: (A)**